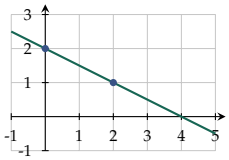
5 One-variable descriptive statistics

Sheet 25 Thu 17 Dec. 2026 *Mean and linearity of the mean*

1. Warm-up · Quick Mean of 12, 15, 9, 14?
2. Core · Method The series 4, 6, 8, 10, 12 has mean 8. What is the mean if 3 is added to each value? If each value is multiplied by 2?
3. Core + · True/false True or false? Justify. The mean of two numbers $a < b$ is always strictly between a and b.
4. Review · Reading In a pie chart of 80 people, one sector represents 25%. How many people does it represent? What is its angle?
5. Thread · Quick Calculate $\gcd(84, 126)$ and deduce the simplest form of $\frac{84}{126}$.

Sheet 26 Mon 4 Jan. 2027 *Median and quartiles*

1. Warm-up · Quick Order 7, 2, 9, 4, 5, 11, 3 then give the median.
2. Core · Method Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give Q_1, the median, Q_3 and the interquartile range.
3. Core + · True/false True or false? Justify. At least half of the values of a series are greater than or equal to its median.
4. Review · Quick Put 0.6, $\frac{5}{8}$ and 61% in increasing order.
5. Thread · Graph → formula The line below represents an affine function f. Determine $f(x)$.



Sheet 27 Tue 5 Jan. 2027 *Standard deviation; effect of adding or removing a value*

1. Warm-up · Quick Calculate $(3-5)^2 + (7-5)^2$.
2. Core · Method Series 2 ; 4 ; 6: mean, variance and standard deviation (approximate value to 10^{-2}).
3. Core + · True/false True or false? Justify. Adding 5 to each value of a series does not change its standard deviation.
4. Review · Construction Let ABC be a triangle, I the midpoint of $[AB]$ and D the reflection of C in I. What type of quadrilateral is $ACBD$?
5. Thread · Quick $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate AN.

Sheet 28 Thu 7 Jan. 2027 *Comparing series*

1. Warm-up · Quick Range of 3, 8, 5, 12?
2. Core · Interpret Two series A and B have the same mean 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which one is more consistent?
3. Core + · Interpret A: median 11, interquartile range 2. B: median 13, interquartile range 6. Compare the two series.
4. Review · Quick Two fair coins are tossed. What is the probability of getting two heads?
5. Thread · Method Programme: choose a number; square it; add 1. Result for -3? Can 0 be obtained? Justify.

Sheet 29 Mon 11 Jan. 2027 *Grouped data: histogram and cumulative frequencies*

1. Warm-up · Quick Give the width and the midpoint of the class $[20, 35)$.
2. Core · Table Classes $[0, 10)$, $[10, 20)$, $[20, 30)$ with frequencies (counts) 4, 6, 10. Give the relative frequencies, then the cumulative increasing relative frequencies.
3. Core + · Quick In a histogram with classes of equal width, two rectangles have heights 3 cm and 6 cm. What is the ratio of their frequencies (counts)?
4. Review · Quick Write 0.00032 in scientific notation.
5. Thread · Quick Solve in $\mathbb{R}$: $2(x + 3) = 5x - 9$.

Sheet 30 Tue 12 Jan. 2027 *Weighted mean, median class; spreadsheet and Python*

1. Warm-up · Quick Calculate $\dfrac{2 \times 8 + 3 \times 12}{5}$.
2. Core · Method Classes $[0, 10)$, $[10, 20)$, $[20, 30)$ with frequencies (counts) 4 ; 7 ; 9. Estimate the mean using the class midpoints; give the median class.
3. Core + · Code reading What does this programme calculate and what value does it give?

```
notes = [12, 15, 9, 14]
s = 0
for n in notes:
    s = s + n
print(s / len(notes))
```

4. Review · Quick An item costing 45,000 Ar is sold at -30%. What is the sale price?
5. Thread · Prove Programme: choose an integer n; add 1; square it; subtract n^2. Show that the result is always an odd number.